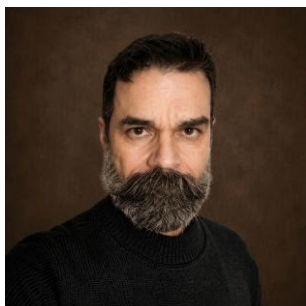




John Hawkins

/ Senior Software Engineer & Kubernetes Expert

Senior Software Engineer & Kubernetes with DevOps experience across software development, Linux infrastructure and production operations. John has worked hands-on with building, maintaining and troubleshooting Kubernetes-based platforms, with recent assignments focused on AWS EKS, GitOps, Terraform, Flux and Kubernetes Operators. He has also supported a migration from an on-premises OpenShift platform to EKS and built deployment pipelines for a local Kubernetes environment.

His technical background includes Linux, Docker, Kubernetes deployments, Argo CD-based delivery flows, Go, Python and Bash. He has also completed training in Helm and Istio. His operational experience covers infrastructure security, monitoring, troubleshooting, disaster recovery and change management.

/ Highlighted assignments

Site Reliability Engineer

Axis

Nov 2024 - Ongoing

I'm on a team building a customer service platform on AWS. EKS is the platform core and I serve as the team Kubernetes expert. The platform is managed as IaC with Terraform in GitHub. I'm implementing a GitOps deployment using the Flux operator, and also handle maintenance, troubleshooting, and documentation. I implement SOC2 remediation findings.

Kubernetes/DevOps/AWS Engineer/Go developer

Bank

Nov 2021 - Jun 2024

The team I belonged to is helping a local bank with their migration from a legacy on premises OpenShift platform to Amazon Web Services hosted EKS Kubernetes. I joined the team as a DevOps/AWS/Kubernetes/Linux engineer and was involved with various aspects of the project, but mainly focused on GitOps and management of deployment to Kubernetes clusters.

Platforms

Kubernetes
Linux
Amazon Web Services
GitHub

Uncategorized

Kubernetes Deployment
Kubernetes Operators
Kubernetes Operator development
YAML Kubernetes

Methods & Processes

GitOps
DevOps

Techniques

Flux
Infrastructure as Code
YAML
Python Scripting

Products

AWS EKS
EKS
Docker
AWS Lambda serverless
Jenkins

Tools

Terraform
Go
Bash Scripting

LinkedIn

<https://www.linkedin.com/in/john-hawkins-488b345/>

/ Highlighted education

Bachelor of Agricultural Science

1983 - 1987

Massey University (Palmerston North)

Kubernetes Certified Application Developer (CKAD) with Tests

2024

Helm Kubernetes Packaging Manager for Developers and DevOps

2024

Istio Hands-On for Kubernetes

2024

/ Relevant projects and assignments

Axis

Site Reliability Engineer

Nov 2024 - Ongoing

In the assignment I am part of the team that is building the customer's new service platform hosted by AWS. EKS - Elastic Kubernetes Service - is the heart of the new platform and my main role is as the team Kubernetes expert.

The entire platform is managed as IaC - infrastructure as code - coded using Terraform and stored on GitHub.

My current focus is implementing a GitOps based deployment strategy, using the GitOps solution Flux operator from Controlplane, but I am also involved with other tasks such as maintenance, troubleshooting and documentation of the platform.

We are also working toward the next SOC2 approval; my role there is mostly as an implementer of findings others have made.

Flux, Kubernetes, Terraform, AWS EKS, GitOps, Kubernetes Operators, Kubernetes Deployment

Bank

Kubernetes/DevOps/AWS Engineer/Go developer

Nov 2021 - Jun 2024

The bank cannot be identified for security and for commercial reasons.

The team I belonged to is helping a local bank with their migration from a legacy on premises OpenShift platform to Amazon Web Services hosted EKS Kubernetes. I joined the team as a DevOps/AWS/Kubernetes/Linux engineer and was involved with various aspects of the project, but mainly focused on GitOps and management of deployment to Kubernetes clusters. My role became much more focused on Kubernetes over the course of the project as AWS Cloud engineers were employed by the Customer at the same time as the provisioning of Cloud infrastructure using Kubernetes Operators played an increasing role in the Customer's DevOps pipeline. My experience as a developer, with infrastructure and with operational support as well as Linux expertise enabled me to fill that role.

My role when I joined the project at the end of 2021 was primarily as an AWS Cloud engineer, with experience working with Kubernetes and DevOps deployment pipelines. As a result of changes in the the project's needs, in part driven by colleagues leaving and in part by evolving project goals, my role became that of a Kubernetes DevOps/GitOps specialist, with a focus on using Kubernetes Operators to provision Cloud infrastructure declaratively. This included coding a new, stand-alone Kubernetes Operator using the Go programming language to manage databases and roles on Postgres servers according to the Bank's own provisioning rules as none of the existing Kubernetes Operators were a good fit.

Aside from the Operator I coded, I worked with the Upbound Crossplane AWS Operator family for provisioning and managing AWS Cloud infrastructure, and the MongoDB Atlas Operator for managing MongoDB Cloud infrastructure. Compared with tools I have used in the past to provision AWS Cloud infrastructure – e.g. Cloud Formation or the AWS CDK – I found the declarative model used in Crossplane to be much simpler and more intuitive.

A further advantage with using Kubernetes Operators to manage the Cloud infrastructure is that the code for this is then in the same format as that for application management, i.e. Kubernetes manifest files. Which means that the infrastructure provisioning code can be handled by the same tools and in the same way as application deployment code, by the same teams and without a need for new skills.

The Kubernetes manifests used to provision the infrastructure were stored in an on-premises BitBucket repository, picked up by a Jenkins pipeline and deployed by ArgoCD into AWS EKS clusters, where the appropriate Kubernetes Operators reconciled the manifests with the existing infrastructure, updating if necessary the Cloud platform infrastructure, to ensure that it was as described. When I joined in late 2021 I was told that the pipeline was a legacy of the on-premises environment and would be soon be replaced by one more appropriate for the new Cloud platform, but unfortunately the old pipeline was still being used when I left the project.

I also ended up taking responsibility for databases – Postgres, MongoDB and Redis – and for disaster recovery including mitigation of ransomware threats to backed up data. Though for disaster recovery I have only been able to recommend a short term (less than 30 days) solution as the Customer is uncertain about the legal requirements related to GDPR rules and planned therefore to hold a workshop to investigate the issue.

Linux, Amazon Web Services, Mongo db, Redis, DevOps, Go, PostgreSQL database, Infrastructure as Code, Kubernetes, YAML, GitOps, Kubernetes Operators, EKS, Crossplane, Kubernetes Deployment, MongoDB Atlas Operator

ABC Labs

DevOps Engineer

Apr 2021 - Sept 2021

I joined the team as a DevOps engineer, with the goal of setting up a DevOps pipeline managed by Jenkins to compile and deploy applications written in Kotlin on a Kubernetes cluster. As it turned out, I ended up spending much of my time working on documentation of the infrastructure as preparation for an upcoming audit and on AWS administration work.

However I did set up a two deployment pipelines managed by a Jenkins controller - one using a Jenkins build node running on a VM, the other using an ephemeral Jenkins build agent running in a Docker container - and Jenkins scripting which was integrated with source code management on both a local GitLab server and public GitHub using webhooks in a development environment. These worked as expected, with code check-in to a repository triggering a new build and deployment of the application on a local Kubernetes cluster. I also created a proof-of-concept pipeline using an ephemeral Jenkins agent hosted on the same Kubernetes cluster, which worked as expected as well. It was clear that ephemeral build agents would be the better solution in a Cloud environment, though it is possible that a Jenkins node running on a VM would be simpler from a maintenance point of view as knowledge of building containers would not be required.

Jenkins, Amazon Web Services, GitHub, Docker, Kubernetes, Jenkins Pipeline, Jenkinsfile

IKEA

Cloud Linux engineer IBES

Feb 2016 - Mar 2021

I was assigned to the project as a Linux expert to help evaluate an existing local on-premises Linux based temporary eCommerce solution ('IBES') for possible global deployment. Time-to-market was the most important factor in our scope as development of the planned long term eCommerce solution had been delayed by several years. In the end we (the evaluation team, which included business analysts, the product owner and developers as well as myself) decided that with some modification the frontend of the existing solution could be used as a base for an Amazon Web Services hosted solution to be rolled out within the time frame the customer required. This would not have been possible with an on-premises based solution.

I worked full time as a Cloud, Linux and scripting expert until the first few markets had been deployed then returned part-time to my previous project as a Tech Lead. I remained in IBES as backup Linux engineer and discussion partner for my full-time colleague and returned to IBES January 2018 full-time when my colleague left.

Nominally I belonged to the platform development sub-team, but in practice much of my time was spent with IT operations because of my technical skills. This suited me as I could identify problems when (in many cases before) they occurred and solve them, often with the help of the application development team.

I created and maintained a number of scripts used for installation (Bash mostly), database backup and management (Python-Lambda), and for making the application server self-healing in some situations we had had issues with, e.g. runaway php-fpm processes and disk capacity issues. I managed a simple build pipeline; originally it included a Jenkins server, but I replaced the Jenkins server with a bash script on a server in the private AWS network as the Jenkins server faced the public network and was therefore a security risk. The rest of the pipeline consisted of a code package delivered by the developers, git clients, CodeCommit repositories and an S3 bucket for the finished installation packages. My installation bundle copied the application package from the S3 bucket during the installation process of the individual server instances.

It wasn't part of my official role, but I also handled most of the change management requests in the customer's change management system as I had the necessary permissions and experience from earlier assignments at the customer site.

Linux, Amazon Web Services, Redis, Bash Scripting, Change Management, Linux Apache MySQL PHP, IT Operations, YAML Ain't Markup Language, AWS Lambda serverless, Python Scripting, DevOps Engineer, MySQL/MariaDB, IT Operations maintenance

IKEA

Tech lead ARDA/IPA project

Oct 2016 - Dec 2017

Part-time tech lead for Oracle products (both databases and application servers).

The project ARDA, which became IPA, was planned as a private cloud in a customer DC to be used to host short lived test environments. It looked good on paper, but in practice turned out to be difficult as one or more components in the stack always needed updating and updates generally broke something else. Cancelled some time after I left.

Project Manager, Linux

IKEA

Chef/Linux engineer - ARDA pilot project

Mar 2015 - Jan 2016

Assigned to the project as a Linux and scripting expert, to evaluate the application Chef for Linux server management. Morphed into a more generic technical role, though with the focus remaining on Linux servers.

Linux, Perl programming

IKEA

Powershell developer

Nov 2014 - Feb 2015

Wrote Powershell scripts to help automate analysis of PCIDSS compliance in the IKEA Windows network.

PowerShell Script

IKEA

Tester; payment systems

Apr 2012 - Oct 2014

The original role was as a tester of IKEA's old Linux based Calypso cash register system and internal payment card handling system iPay, but I ended up working with production support of Calypso and with the new Windows based iPOS cash register system.

Most of the testing involved webservices calls to servers so I wrote a set of Perl scripts to automate the execution and analysis of test cases.

Perl programming

IKEA

Analyst and project lead/coordinator

Apr 2011 - Mar 2012

Analyst and project lead/coordinator for a one-man project to replace the Windows Reflection client with a browser based client world wide, saving the customer around 6 million € yearly in licence costs.

Involved analysis of licence data, coordination with the various country IT managers and change management.

Change Management

Sony Ericsson

Requirements management tools support

Apr 2008 - Mar 2010

User and application support plus application development using the built-in DOORS scripting language.

Sony Ericsson

Requirements manager

Aug 2006 - Mar 2008

Requirements management using the DOORS requirements management application. In addition to my regular requirements manager work I also did a full audit of the requirements database to identify poorly written or ambiguous requirements.

Requirement Manager

Sony Ericsson

Test of internet security

Aug 2003 - Jul 2005

Nominally the role was for Sony Ericsson's mobile phone client stack and encryption of Internet traffic, but I also ended up administering Apache web servers (Linux) and a WAP server (Solaris) as well as writing a lot of Perl code that automated capture and analysis of network traffic.

Linux, Perl programming, Bash Scripting, Quality assurance, Apache Web Server

Capgemini Sweden

Linux engineer

Sept 2002 - Mar 2003

Internal project. Linux administration and scripting.

Installation and configuration of the 'Big Brother' open source system monitoring application for internal use. The licence cost of the existing system - Computer Associates Unicenter - was too high so Big Brother was installed while a longer term solution using Microsoft's own System Management Server was under development.

I was responsible for central server installation and configuration, installation on Linux servers and the building + installation of the Solaris client package.

Linux, Perl programming, Bash Scripting, Solaris/Unix

Capgemini Sweden

Linux engineer

Aug 2001 - Aug 2002

Internal project

Linux engineer for a pilot project for a commercial Sendmail product to provide email and calendar services for Capgemini Europe. Used both Linux and Solaris servers.

Sendmail, Linux

L M Ericsson AB

Technical Consultant

Jun 2000 - Jun 2001

Development of a backup, restore and disaster recovery solution for radio base stations.

Single person team, reporting to the project manager.

Scripts were written in Perl and Bash, and a commercial disk mirroring solution was used as Solaris did not have native support for this at the time.

Sun Solaris, Perl programming, Bash Scripting

Various

Local support and administration for various companies

Aug 1998 - Jun 2000

I was part of the support and IT administration team that worked with a number of contracts for IT support and administration that Capgemini had with companies in and around Helsingborg. Mostly Microsoft Windows, but some Linux also.

/ Past employments**Knowit Cloud AB**

Apr 2021 - Ongoing

Capgemini Sweden, Malmö Sweden

Aug 1998 - Mar 2021

Ericsson Eurolab Deutschland, Nürnberg, Germany

Aug 1996 - Jul 1998

Medea Systems AB, Täby Sweden

Aug 1994 - May 1996

Foraco AB, Falun Sweden

Jan 1992 - Aug 1994

Stora Skog, Falu Skogsförvaltning, Falun Sweden

Aug 1988 - Dec 1991

NZFP Forests, Kinleith site, Tokoroa New Zealand

Apr 1987 - Feb 1988

/ Education**Bachelor of Agricultural Science (Bachelors)**

1983 - 1987

Massey University (Palmerston North)

/ Courses & Certifications**Kubernetes Certified Application Developer (CKAD) with Tests**

2024

Helm Kubernetes Packaging Manager for Developers and DevOps

2024

Istio Hands-On for Kubernetes	2024
Go - The Complete Guide	2024
Working with Microservices in Go (Golang)	2024
Certified SAFe 4 DevOps Practitioner Scaled Agile	2019
AWS Certified Sysops Administrator –Associate Amazon Web Services	2018

/ Language skills

English	Native or bilingual proficiency
Swedish	Full professional proficiency
German	Elementary proficiency